

Study guide for midterm exam

GRAD-E1394

October 21, 2024

*Note on the study guide: These problems and tips will give you an indication how to study for the exam. They do **not** provide a comprehensive list of problems on the midterm, there may be different types of questions and different topics on the exam. On the first page of the midterm, you will find general instructions that read similar to the following:*

When you provide the solution, always show how you reached your answer and clearly state your assumptions (unless stated otherwise). You do not need to start from first principles, but your answer should show be sufficiently detailed that we can understand how you derived your solution from the premise of the problem. Stating how a solution can be found will give you partial credit. Give numerical answers in their simplest form. Appropriately label the axes of your graphs and add a legend or labels to help the reader understand what is shown.

1 Study tips

The midterm will cover Sessions 1-6 (by lecture notes, it will also cover part of Lecture 7), and the problem sets (emphasis on theoretical part but also conceptual things from the practical part). The following types of problems may appear on the midterm:

- True or false questions
- Multiple choice questions
- Mathematical derivations
- Calculations
- Conceptual questions that can be answered with a mathematical expression, text, or graph.

Study tips:

- Study the lecture notes and problem sets carefully.

- Practice modifications of the problem sets and practice problems, and also think about what other types of problems may be asked for.
- Use Zhang et al. textbook & exercises as a resource (exercises are often more advanced)
- Compute gradients of different functions. Also familiarize yourself of the derivatives of common activation functions.
- Be familiar with how neural networks and backpropagation are formulated in the lecture and problem sets.
- How different convolutional filters activate on an image.
- Convolution operation in 1 and 2 dimensions.
- Don't constrain yourself to the areas mentioned in the study guide.
- This class builds on the Machine Learning and Maths for Data Science courses. We expect you to have a working knowledge of important concepts covered there that are needed for deep learning.

2 Example questions

Solutions are on the last page.

1. When tuning the hyperparameters with a grid search for 5 different values of the learning rates and batch size of 16 and 32, and you compare your model errors in a 5-fold cross validation setting where for each fold you do the hyperparameter tuning, how many times will the model be trained?
2. When training a neural network over 50 epochs with 5,000 training samples and using a batch size of 10, how often is all the data run through the model?
3. You are given the neuron h , expressed as

$$h(\mathbf{x}) = g \left(b + \sum_{j=1}^d w_j x_j \right) \quad (1)$$

with the input data $\mathbf{x} \in \{0,1\}^d$, bias $b \in \mathbb{R}$, weight $\mathbf{w} \in \mathbb{R}^d$, and the activation function $g(z) = \frac{1}{1+e^{-z}}$.

For 2-dimensional binary input data $\mathbf{x} = (x_1, x_2)$, can the neuron learn the following problem: $AND(x_1, x_2)$? Provide your answer and graph the problem with appropriate description of the graph and labels. If applicable, draw the decision boundary that the neuron can learn.

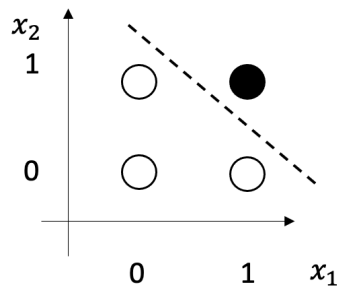
4. True or false? Circle the answer (no need to provide derivations).
 - (a) Can a neural network with one hidden unit and one output unit and sigmoid activation functions learn the $XOR(x_1, x_2)$ problem? **True** / **False**
 - (b) Since convolutions are translation equivariant, for multi-layer convolutional neural networks there is little use for applying translation in data augmentation. **True** / **False**
 - (c) Recurrent Neural Networks (RNNs) can handle sequences of different lengths. **True** / **False**
5. You find that your training error becomes very low during training, however, your generalization error stays high.
 - (a) Are you seeing a case of overfitting or underfitting?
 - (b) Name a potential issue that may be causing that problem and two common deep learning techniques to mitigate the problem.
6. You discover that your convolutional neural network kernel has learned the following weights (the operation is implemented as a cross-correlation)

$$K = \begin{bmatrix} -0.98 & 0.04 & 0.78 \\ -0.87 & -0.05 & 0.92 \\ -0.99 & -0.01 & 0.96 \end{bmatrix}. \quad (2)$$

Describe what the kernel is filtering for in one or two sentences.

3 Solutions

1. The model is trained $5 \times 2 = 10$ times for each run of the 5-fold cross-validation. The solution is therefore: $10 \times 5 = 50$.
2. The solution is 50.
3. The single neuron can learn linearly separable problems, which means that it can learn the AND function. In the graph, black indicates $x_j = 1$ and white indicates $x_j = 0$. The dashed line is the decision boundary.



4. True or false? Circle the answer (no need to provide derivations).
 - (a) Can a neural network with one hidden unit and one output unit and sigmoid activation functions learn the $XOR(x_1, x_2)$ problem? **Solution: False.** It would need to have more than one hidden unit. There is no need to have multiple hidden layers.
 - (b) Since convolutions are translation equivariant, for multi-layer convolutional neural networks there is little use for applying translation in data augmentation. **Solution: False.** For the higher layers, which can be understood to process the image at lower resolution, this is still important.
 - (c) Recurrent Neural Networks (RNNs) can handle sequences of different lengths. **Solution: True**
5.
 - (a) The model is overfitting.
 - (b) Use regularization techniques to reduce the model complexity:
 - Weight sharing (which greatly reduces the number of parameters)
 - Dropout
 - Weight decay
 - Sparsity in the hidden units
6. The kernel is identifying a vertical edge from dark to bright (left to right).